

Sanjith Menon (He/Him/His)

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Education

University of California, Los Angeles (UCLA)

B.S., Mechanical Engineering, GPA: 3.958

Los Angeles, CA

Sep. 2022 – Jun. 2026

Honors/Awards: Magna Cum Laude, Dean's Honors List, ALD|PES Honors Society, Tau Beta Pi Engineering Honors Society

Stanford University

M.S., Mechanical Engineering

Stanford, CA

Sep. 2026 – Jun. 2028

Professional Experience

Harbinger Motors – Product Engineering Intern, Powertrain

Jun. 2026 – Aug. 2026

- Designed flywheel loading system supporting 1400+ lbs. to align and assemble inertial loads for park lock test rig.
- Analyzed fasteners in electric drive unit and recommended changes to patch lock and install torque for reliability.
- Fabricated oil temperature sensor harness for end-of-line coolant oil check to reduce faults in production units.
- Performed vibration (NVH) testing of oil pumps for electric drive unit and identified areas of interest for future test.

Rivian Automotive – Manufacturing Engineering Intern

May. 2025 – Dec. 2025

- Owned end-to-end design (CATIA), validation, and deployment of custom hood jig fixtures that eliminated 100% of bracket deformation and cut the number of distinct hood jigs in paint process by 25%.
- Led implementation, installation, and commissioning of \$200K e-coat air blower system reducing defects by 19% and of custom feather duster station reducing defects by 24%.
- Performed a data-driven root cause analysis of skid extension station faults and implemented and validated design changes, reducing faults by 99%.
- Developed proximity switch protection covers to minimize equipment damage and downtime — patent pending.

Southern California Edison – Mechanical Engineering Intern

Jun. 2024 – Sep. 2024

- Developed an interactive tool for the Automated Demand Response program that achieved process improvement—reducing energy load shed calculation time from ~1 hour to 5 minutes, automating data visualization, and eliminating 90% of redundant processes in reporting, technical reviews, and data analytics.
- Inspected multiple sites for compliance with SCE commercial customer program requirements.
- Shadowed engineers evaluating emerging technologies and managing programs for energy efficiency and grid load management by performing technical reviews and operating Technology Test Center lab and Heat Pump Water Heater lab.

Research Experience

UCLA Structures-Computer Interaction Lab – Hidden Fruit Detection Researcher

Jan. 2025 – Apr. 2025

- Co-authored research paper on hidden fruit detection via sensor fusion.
- Performed rapid prototyping for sensor module combining NIR, FLIR, RGBD cameras.
- Collected thermal imagery data of occluded fruits via various cameras/sensors while heating fruits to train identification model.

UCLA Structures-Computer Interaction Lab – LLM Robotics Researcher

Oct. 2024 – Dec. 2024

- Authored 4-page technical report detailing control and operation of robotic arm.
- Designed and fabricated holder attachments for the robotic arm, factoring in design considerations such as strengthening failure prone sites.
- Programmed Python functions for control and manipulation of robotic arm to be added to library of tools for LLM.
- Explored robot controlling software ROS in Ubuntu.

UCLA Structures-Computer Interaction Lab – Deployable Structures Researcher

Feb. 2024 – Jun. 2024

- Co-authored research paper on shrinkable thermoplastic sheets (Shrinky-Dink) and their applications in heat-activated composite structures.
- Conducted experiments to convert flat designs to 3D shapes via heat application and collect temperature and time data.

- Analyzed temperature data during deformation of material using MATLAB to generate graphs needed for visualization of heating process to be included in research paper.
- Hands-on experience in 3D printing designs, laser cutting material, epoxy adherence to create composite material, heating the composite, analyzing data, performing literature review.

Garrick Institute for the Risk Sciences – *Wildfire Risk Mitigation Optimization Researcher* *Dec. 2023 – Jun. 2024*

- Studied deep learning and machine learning to optimize risk mitigation strategies for wildfire emergencies in electric transmission networks.
- Utilized GIS software to model transmission lines versus high wildfire risk areas geospatially to find correlations.

Publications

Mrunmayi Mungekar, **Sanjith Menon**, M. Ravi Shankar, M. K. Jawed, Advanced Engineering Materials 2026, 0, e202502378. <https://doi.org/10.1002/adem.202502378>

Projects and Project Team Experience

Automatic Trash-Catching Robot

Jul. 2026 – Present

- Building a trash-catching robot where the trashcan is mounted on a linear rail and 2D camera feedback triggers Arduino stepper motor control based on computer vision motion tracking to catch a crumpled tissue paper.
- Performed motor speed analysis, mechanical and electrical circuit design, and constructed a finite state machine-based control programming structure inside a ROS2 environment.
- Learning about computer vision methods of motion trajectory tracking and data transfer between Arduino and PC.

UCLA FSAE Bruin Formula Racing – *Powertrain Engineer*

Oct. 2022 – Jun. 2026

- Designed ducting system for airflow enhancement and waterproofing of battery pack cooling and integrated it into powertrain system — achieved over 200% increase in air velocity during vehicle motion and passed rain test.
- Performed CFD analysis (Siemens) to validate cell segment temperatures and duct flow rates.
- Analyzed fan curves and cooling performance via MATLAB models based on heat transfer and fluid mechanics.
- Designed, manufactured, assembled motor test bench using machine shop skills to validate functionality of motor.

Autonomous Burger Assembly/Delivery Robot – *Mechatronics Engineer*

Jan. 2026 – Jun. 2026

- Built and programmed an autonomous differential-drive robot to perform LiDAR-based course navigation, computer vision-guided customer recognition, and mechanical manipulation for burger assembly that ranked first in the capstone class.
- Implemented a non-blocking Python Finite State Machine (FSM) utilizing ROS 2 to orchestrate robot operation while prioritizing critical task timing and running a PID-based firmware control system.

Robotic Arm Project – *Project Team Member*

Jun. 2023 – Aug. 2023

- Created CAD parts and subassemblies in SolidWorks for 6 DOF robotic manipulator complete with motor models.
- Created engineering drawings with an emphasis on GD&T standards for parts and subassemblies of robotic arm, as well as overall robotic arm drawing complete with BOM.
- Demonstrated hypothetical operation of robotic arm through presentation and 100-page technical report.

MATLAB Ant Colony Simulation – *Project Team Member*

Jun. 2023 – Aug. 2023

- Programmed using MATLAB to simulate an ant colony's behavior in finding food using pheromone trails.
- Created multiple functions and iteratively conducted test simulations to achieve ideal result.
- Programmed the simulation to record each frame and created a video for demonstration of functionality.
- Produced a final report explaining the algorithm and results.

Leadership

Tau Beta Pi Engineering Honors Society – *Internal Vice President*

Feb. 2025 – Jun. 2026

- Managed committees focused on student body support like tutoring, member coordination, project chairs, social.

Tau Beta Pi Engineering Honors Society – *Project Chair*

June. 2023 – Feb. 2025

- Organized engineering events like innovative challenges (cardboard boat race, cardboard chair competition), academic competitions (Engineering Bowl), and large-scale events (banquet, TBProm).

Skills

Engineering Skills: CATIA CAD, SolidWorks CAD, MATLAB, Altair Inspire FEA, Ansys FEA, SolidWorks FEA, Siemens CFD, Breadboard/circuit design, Arduino, Sawyer Robot

Computer Skills: Python, Java, JavaScript, HTML 5, C++, Microsoft Excel/Google Sheets, VBA, Microsoft Copilot, Power BI, Ubuntu, ROS2

Machine Shop Skills: 3D Printing, Laser Cutter, Mill, Lathe, Bandsaw, Drill Press, Dremel, Soldering